Year 1 AS Pure Maths

Chapter 7

Algebraic methods

Practice Booklet

Contents

7.1 Algebraic fractions ..... book p. 55
7.2 Dividing polynomials ..... book p. 55
7.3 The factor theorem ..... book p. 56
7.4 Mathematical proof ..... book p. 58
7.5 Methods of proof ..... book p. 59
Problem solving: Set A ..... book p. 60
Problem solving: Set B ..... book p. 61
Selected exam question bank questions .....
Answers .....

7.1 Algebraic fractions

Practice Book, printed page 55.

1. Simplify these fractions:

a. $\dfrac{6x^4 + 8x^3 - 4x^2}{2x}$

b. $\dfrac{12x - 24x^2 + 9x^3}{3x}$

c. $\dfrac{8x^6 - 4x^3 + 2x^4}{-2x^2}$

Hint

Divide each term in the numerator by the denominator.

2. Write each of these fractions in their simplest form:

a. $\dfrac{x^2 - 9}{x - 3}$

b. $\dfrac{x^2 - 3x + 2}{x - 2}$

c. $\dfrac{2x^2 + 5x - 3}{x + 3}$

Hint

To simplify an algebraic fraction, factorise the numerator and denominator where possible and cancel common factors.

3. Write each of these fractions in their simplest form:

a. $\dfrac{2x^2 - 3x - 2}{x^2 + x - 6}$

b. $\dfrac{3x^2 + x - 2}{2x^2 + 5x + 3}$

c. $\dfrac{6x^2 + 11x - 10}{2x^2 - 3x - 20}$

4. E/P Show that $\dfrac{15x^2 - 16x - 15}{3x^2 - 5x}$ can be written in the form $A + \dfrac{B}{x}$ where A and B are integers to be found. (3 marks)

5. E/P $\dfrac{x^2 - 2x + k}{x^2 + 2x - 3} = \dfrac{x - 5}{x - 1}$ where k is a constant. Work out the value of k. (3 marks)

6. E/P $\dfrac{4x^2 + 8x - 5}{6x^2 - 7x + k} = \dfrac{2x + 5}{3x - 2}$ where k is a constant. Work out the value of k. (4 marks)

7. E/P $\dfrac{6x^3 - 2x^2 - 4x}{2x^2 - 7x + 5} = \dfrac{ax(bx + c)}{px + q}$, where a, b, c, p and q are constants.
Work out the values of a, b, c, p and q. (5 marks)

7.2 Dividing polynomials

Practice Book, printed page 55–56.

1. Write each polynomial in the form $(x \pm p)(ax^2 + bx + c)$ by dividing:

a. $2x^3 - 5x^2 + 8x - 5$ by $(x - 1)$

b. $3x^3 + 8x^2 + 3x - 2$ by $(x + 2)$

c. $2x^3 + x^2 - 17x - 12$ by $(x - 3)$

d. $4x^3 + 13x^2 - 11x + 4$ by $(x + 4)$

Hint

You can use long division to divide a polynomial by $(x \pm p)$, where p is a constant.
For example:

$$x - 1 \overline{)\,2x^3 - 5x^2 + 8x - 5}\quad \begin{matrix} 2x^2 \\ \\ \underline{2x^3 - 2x^2} \end{matrix}$$

and so on

2. Divide:

a. $3x^4 + 8x^3 - x^2 - 13x - 6$ by $(x + 2)$

b. $4x^4 - 8x^3 + x^2 - x - 2$ by $(2x + 1)$

c. $9x^4 - 3x^3 - 17x^2 + 13x - 2$ by $(3x - 2)$

d. $4x^4 - 12x^3 - 5x^2 + 15x + 9$ by $(2x - 3)$

Hint

The answer you obtain following the division is called the **quotient**.

3. Divide:

a. $2x^3 + 6x^2 - 4$ by $(x + 1)$

b. $3x^3 + 7x^2 + 18$ by $(x + 3)$

c. $4x^3 - 11x - 10$ by $(x - 2)$

d. $2x^3 + 7x^2 + 75$ by $(x + 5)$

Hint

Write the polynomial in part **a** as $2x^3 + 6x^2 + 0x - 4$ before dividing.

4. Find the remainder when:

a. $x^3 + 3x^2 + 5x - 8$ is divided by $(x + 4)$

b. $2x^3 - 5x^2 + 12x - 20$ is divided by $(x - 3)$

c. $3x^3 + 2x^2 - 40x + 45$ is divided by $(x + 5)$

Hint

If there is a remainder, then the linear expression $(x \pm p)$ is not a factor.
The polynomial can be written as $(x \pm p)(ax^2 + bx + c) + r$ where r is the remainder.

5. E Find the remainder when $-15x^3 + 26x^2 - 13x + 5$ is divided by $(5x - 2)$. (2 marks)

6. E/P $f(x) = 6x^3 - 13x^2 - 13x + 30$

a. Find the remainder when $f(x)$ is divided by $(x + 3)$. (2 marks)

b. Given that $(x - 2)$ is a factor of $f(x)$, factorise $f(x)$ completely. (4 marks)

7. E/P $f(x) = 2x^3 + 3x^2 - 4x + k$ where k is a constant.
Given that $(x + 3)$ is a factor of $f(x)$:

a. find the value of k (2 marks)

b. express $f(x)$ in the form $(x + 3)(ax^2 + bx + c)$ where a, b and c are constants (2 marks)

c. show that $f(x) = 0$ has exactly one real solution. (2 marks)

8. E/P $f(x) = 3x^3 + 10x^2 - 8x - 5$

a. Find the remainder, r, when $f(x)$ is divided by $(x - 2)$. (2 marks)

b. Express $f(x)$ in the form $(x - 2)(ax^2 + bx + c) + r$, where a, b and c are constants. (4 marks)

9. E/P $f(x) = 10x^3 - 29x^2 + 4x + 15$

a. Given that $(x - 1)$ is a factor of $f(x)$, express $f(x)$ in the form $(x - 1)(ax^2 + bx + c)$, where a, b and c are constants. (2 marks)

b. Hence factorise $f(x)$ completely. (4 marks)

c. Write down all the solutions to the equation $f(x) = 0$. (2 marks)

7.3 The factor theorem

Practice Book, printed page 56–58.

1. Use the factor theorem to show that:

a. $(x + 1)$ is a factor of $2x^3 + 7x^2 - 5$

b. $(x + 2)$ is a factor of $x^3 + 4x^2 + 3x - 2$

c. $(x - 3)$ is a factor of $2x^3 - 3x^2 - 7x - 6$

d. $(x - 4)$ is a factor of $x^4 - 3x^3 - 15x - 4$

Hint

The **factor theorem** states that if $f(x)$ is a polynomial, then:

- if $f(p) = 0$ then $(x - p)$ is a factor of $f(x)$
- if $(x - p)$ is a factor of $f(x)$ then $f(p) = 0$

2. Use the factor theorem to show that the linear expression is a factor of the polynomial $f(x)$ and factorise $f(x)$ completely:

a. $(x - 2)$, $2x^3 + x^2 - 13x + 6$

b. $(x + 3)$, $2x^3 + 17x^2 + 38x + 15$

c. $(x - 1)$, $6x^3 - x^2 - 11x + 6$

d. $(x + 4)$, $15x^3 + 61x^2 - 2x - 24$

Hint

When you have used the factor theorem to show the linear expression is a factor, you can use long division to find the quadratic factor. Factorise the quadratic factor to write the polynomial as a product of three linear factors.

3. Fully factorise each expression:

a. $x^3 + 2x^2 - 21x + 18$

b. $2x^3 + 13x^2 + 13x - 10$

c. $3x^3 + 2x^2 - 41x - 60$

Hint

Try values of p in each expression for $f(x)$, e.g. $p = -1, 1, 2, 3, \ldots$ until you find $f(p) = 0$. Then use the factor theorem to deduce that $(x - p)$ is a factor of $f(x)$.

4. For each of the following polynomials,
i fully factorise each polynomial $f(x)$.
ii Hence sketch the graph of $y = f(x)$.

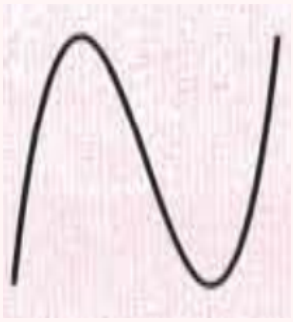
a. $2x^3 - 11x^2 + 5x + 18$

b. $2x^3 - 3x^2 - 39x + 20$

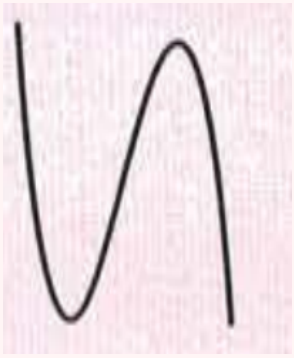
c. $6x^3 + 37x^2 + 50x - 21$

Hint

To sketch the graph, you need to identify the points where the curve crosses the axes.
Set $x = 0$ to find the y-intercept and $y = 0$ to find the x-intercepts.
The general shapes of cubic graphs are:



if the coefficient of x^3 is positive



if the coefficient of x^3 is negative

← Section 4.1

5. E/P $f(x) = 6x^3 - 17x^2 - 15x + 36$
Given that $(x - 3)$ is a factor of $f(x)$, find all the solutions to $f(x) = 0$. (5 marks)

6. E/P $f(x) = 9x^3 + 24x^2 - 44x + 16$

a. Use the factor theorem to show that $(x + 4)$ is a factor of $f(x)$. (2 marks)

b. Hence show that $f(x)$ can be written in the form $f(x) = (x + 4)(px + q)^2$, where p and q are integers to be found. (4 marks)

7. E $f(x) = 2x^3 - 3x^2 - 5x + 6$. Factorise $f(x)$ completely. (5 marks)

8. E/P $g(x) = x^3 + 2x^2 - 19x + k$
Given that $(x + 1)$ is a factor of $g(x)$,

a. show that $k = -20$ (2 marks)

b. express $g(x)$ as a product of three linear factors. (3 marks)

c. Sketch the curve with equation $y = x^3 + 2x^2 - 19x - 20$, indicating the values where the curve crosses the x-axis and the y-axis. (4 marks)

9. E/P $p(x) = 25x^3 + 55x^2 - 56x + 12$

a. Use the factor theorem to show that $(x + 3)$ is a factor of $p(x)$. (2 marks)

b. Fully factorise $p(x)$. (3 marks)

c. Hence show that there are exactly two real roots of the equation $p(x) = 0$. (2 marks)

7.4 Mathematical proof

Practice Book, printed page 58–59.

1. Prove that $(x-4)(x+6)(2x+3) \equiv 2x^3+7x^2-42x-72$

Hint

The identity symbol $\equiv$ means 'is equivalent to'. This means the expressions are equal for all values of x. To prove an identity you should:

- start with the expression on one side of the identity
- algebraically manipulate the expression until it matches the other side
- show every step in your algebraic working.

2. Prove that the triangle with vertices at $P(10, 14)$, $Q(-6, 2)$ and $R(12, 8)$ is right angled.

Hint

In a mathematical proof you must:

- state any information or assumptions you are using
- clearly show every step of your working
- make sure each step follows logically from the previous step
- write a statement of proof at the end of your working.

You could find the lengths of the sides in this triangle and use Pythagoras' theorem, or use gradients of line segments. ← Section 5.4

3. Prove that $y = kx$ is a tangent to the curve with equation $y = 4x^2 - 5x + 4$ when $k = 3$ and $k = -13$.

Hint

If a line is a tangent to a curve, then there is only one point of intersection. Solve the equations simultaneously and use the discriminant. ← Section 2.5

4. Prove that the points $A(-2, 3)$, $B(2, 1)$ and $C(14, -5)$ are collinear.

Hint

If points are collinear, they lie on the same straight line. ← Section 5.1

5. E/P Prove that $x^2 - 6x + 10 > 0$ for all values of x. (3 marks)

6. E/P Prove that $(x+y)^2 - (x-y)^2 \equiv 4xy$ (3 marks)

7. E/P Prove that the difference between the squares of two consecutive odd numbers is a multiple of 8. (4 marks)

8. E/P Prove that the difference between the cube and the square of an odd number is even. (4 marks)

9. E/P The equation $x^2 + (k-3)x + (3-2k) = 0$ where k is a constant has no real roots. Prove that k satisfies $-3 < k < 1$. (6 marks)

7.5 Methods of proof

Practice Book, printed page 59.

1. $f(n) = n^2 + n + 17$ for $n \in \mathbb{N}$.

a. Prove that for $5 \leqslant n \leqslant 10$, $f(n)$ is prime.

b. Find a counter-example to prove that $f(n)$ is not prime for all positive integers, n.

Hint

You can prove a mathematical statement is not true by a **counter-example**. A counter-example is one example that does not work for the statement. One example is sufficient to disprove the statement.

2. Prove that the following statement is not true.
'The difference of two prime numbers is always even.'

3.

a. Prove that for positive values of x, $(1 + 2x)^2 > 1 + 4x^2$

b. Find a counter-example to prove that the inequality $(1+2x)^2 > 1+4x^2$ is not true for all values of x.

4. A student is trying to prove that $\dfrac{x(x+2)^2 + x}{x}$, is a positive numbers of all real values of x.
The student writes:

$$\frac{x(x+2)^2 + x}{x} \equiv \frac{x(x+2)^2}{x} + \frac{x}{x} \equiv (x+2)^2 + 1$$

As $(x + 2)^2 \geqslant 0$ for all values of x then $(x + 2)^2 + 1 > 0$ for all values of x.

a. Identify the error made in the proof.

b. State the value of x for which the statement is not true.

5. E/P Use a counter-example to show that the following statement is false.
'If you add 2 to a number and square it, the result is always greater than the square of the number.'
(2 marks)

6. E/P Prove that $(2n + 1)^2 > 2^{n+1}$ for all positive integers less than 7. (3 marks)

7. E/P Amir claims that $x^2 \geqslant x$.
Determine whether Amir's claim is always true, sometimes true or never true, justifying your answer.
(2 marks)

8. E/P Prove that $(x + 5)^2 \geqslant 4x + 9$ for all real values of x. (3 marks)

9. E/P Prove that if n is an integer and $2 \leqslant n \leqslant 7$, then $n^2 + 2$ is not divisible by 4. (3 marks)

Problem solving: Set A

Practice Book, printed page 60.

Bronze

$p(x) = x^3 - 9x^2 + 2x + 48$

a. Find the remainder, r, when $p(x)$ is divided by $(x - 4)$. (2 marks)

b. Hence write $p(x)$ in the form $(x - 4)(x^2 + ax + b) + r$ where a, b and r are constants. (2 marks)

c. Given that $(x + 2)$ is a factor of $p(x)$, factorise $p(x)$ completely. (4 marks)

d. Prove that if n is an integer and $3 < n < 8$, then $n^3 - 9n^2 + 2n + 48 < 0$ (3 marks)

Silver

$p(x) = 2x^3 - 15x^2 + 32x + 20$
Given that $(2x + 1)$ is a factor of $p(x)$,

a. use long division to express $p(x)$ in the form $(2x + 1)(x^2 + ax + b)$, where a and b are constants to be found. (3 marks)

b. Hence show that $p(x) = 0$ has only one real root. (4 marks)

c. Prove that $2x^3 - 15x^2 + 32x + 20 > 0$ for all real positive values of x. (3 marks)

Gold

$p(x) = 9x^3 - 6x^2 - 20x - 8$
Given that $p(x) = (x + d)(ax^2 + bx + c)$, where a, b, c and d are integers,

a. find the values of a, b, c and d. (6 marks)

b. Prove that $p(x) = 0$ has exactly two real roots. (3 marks)

c. Sketch the curve with equation $y = 9x^3 - 6x^2 - 20x - 8$, indicating the values where the curve cuts the x-axis and the y-axis. (4 marks)

d. Hence or otherwise, use set notation to write down the solutions to the inequality $9x^3 - 6x^2 < 20x + 8$ (3 marks)

Problem solving: Set B

Practice Book, printed page 61.

Bronze

$f(x) = x^3 - x^2 + px + q$ where p and q are integers.
Given that $(x + 1)$ is a factor of $f(x)$,

a. show that $q - p = 2$. (3 marks)

Given that $(x + 3)$ is also a factor of $f(x)$,

b. show that $q - 3p = 36$. (3 marks)

c. Hence find the value of p and the corresponding value of q. (2 marks)

d. Factorise $f(x)$ completely. (2 marks)

Silver

$f(x) = 2x^3 - x^2 + px + q$ where p and q are integers.
Given that $(x + 2)$ is a factor of $f(x)$,

a. show that $q - 2p - 20 = 0$. (3 marks)

Given that $(x - 3)$ is also a factor of $f(x)$,

b. find the value of p and the corresponding value of q. (5 marks)

c. Factorise $f(x)$ completely. (2 marks)

Gold

$f(x) = x^3 + (p + 4)x^2 + 8x + q$ where p and q are integers.
Given that $(x - 2)$ is a factor of $f(x)$,

a. show that $4p + q + 40 = 0$. (3 marks)

Given that $(x + p)$ is also a factor of $f(x)$, and that $p > 0$,

b. show that $4p^2 - 8p + q = 0$. (3 marks)

c. Hence find the value of p and the corresponding value of q. (5 marks)

d. Factorise $f(x)$ completely. (2 marks)

Selected exam question bank questions

Question 30

a. Prove that when p and q are positive real numbers with $p > q$,
$\sqrt{p^2 - q^2} > p - q$ (3 marks)

b. Prove by counter-example that this is not true for all values of p and q. (1 mark)

Question 34

a. Use a counter-example to show that the following statement is false.
'$5^x \geqslant 2^x$ for all values of x' (2 marks)

b. Alison chooses two consecutive odd numbers. She squares them, adds the results, then halves the total. Prove that the result is always an odd number.
For example, $\frac{1}{2}(9^2 + 11^2) = 101$ (3 marks)

Question 41

$f(x) = 6x^3 + 5x^2 - 3x - 2$

a. Factorise $f(x)$ completely. (6 marks)

b. Show that $\dfrac{6x^3 + 5x^2 - 3x - 2}{2x^3 + 3x^2 + x}$ can be written in the form $A + \dfrac{B}{x}$, where A and B are integers to be found. (3 marks)

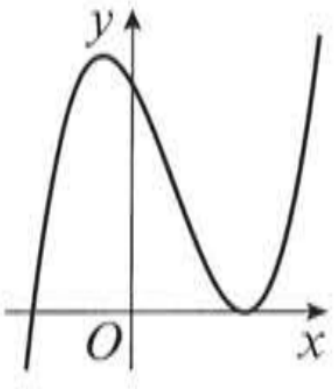
Question 57

$g(x) = 4x^3 - 16x^2 - 35x + 147$

a. Using the factor theorem, explain why $g(x)$ is divisible by $(x + 3)$. (2 marks)

b. Hence show that $g(x)$ can be written in the form $g(x) = (x + 3)(ax + b)^2$ where a and b are integers to be found. (4 marks)

The diagram shows a sketch of part of the curve with equation $y = g(x)$.



c. Use your answer to part **b** and the sketch to deduce the values of x for which:
i $g(x) \leqslant 0$ **ii** $g(2x) \leqslant 0$ (3 marks)

Question 69

$f(x) = (x^2 + k)(2x + 5) + 7$ where k is a constant.

a. Write down the remainder when $f(x)$ is divided by $(2x + 5)$. (1 mark)

Given that $(x - 1)$ is a factor of $f(x)$,

b. prove that $k = -2$ (3 marks)

c. Find all the solutions to $f(x) = 0$. (5 marks)

Question 75

$f(x) = 2x^3 + kx^2 - 24x + 45$ where k is a constant.

Given that $(x - 5)$ is a factor of $f(x)$,

a. show that $k = -7$ (2 marks)

b. factorise $f(x)$ completely. (4 marks)

Given that $g(y) = 2(5^{3y}) - 7(5^{2y}) - 24(5^y) + 45$,

c. find the values of y that satisfy $g(y) = 0$, giving your answers to 2 decimal places where appropriate. (3 marks)

Question 83

$f(x) = x^3 - 2x^2 + x + 18$

a. Show that $(x + 2)$ is a factor of $f(x)$. (2 marks)

b. Hence show that -2 is the only real root of $f(x)$. (4 marks)

c. Prove that $f(x) \geqslant 18$ for all $x \geqslant 0$. (3 marks)

Answers

7.1 Algebraic fractions

1. **a** $3x^3 + 4x^2 - 2x$ **b** $4 - 8x + 3x^2$ **c** $-4x^4 + 2x^3 - x^2$

2. **a** $x + 3$ **b** $x - 1$ **c** $2x - 1$

3. **a** $\frac{2x+1}{x+3}$ **b** $\frac{3x-2}{2x+3}$ **c** $\frac{3x-2}{x-4}$

4. $\frac{(3x-5)(5x+3)}{x(3x-5)} = \frac{5x+3}{x} = 5 + \frac{3}{x}$

5. $k = -15$

6. $k = 2$

7. $a = 2$, $b = 3$, $c = 2$, $p = 2$, $q = -5$

7.2 Dividing polynomials

1. **a** $(x - 1)(2x^2 - 3x + 5)$ **b** $(x + 2)(3x^2 + 2x - 1)$
c $(x - 3)(2x^2 + 7x + 4)$ **d** $(x + 4)(4x^2 - 3x + 1)$

2. **a** $3x^3 + 2x^2 - 5x - 3$ **b** $2x^3 - 5x^2 + 3x - 2$
c $3x^3 + x^2 - 5x + 1$ **d** $2x^3 - 3x^2 - 7x - 3$

3. **a** $2x^2 + 4x - 4$ **b** $3x^2 - 2x + 6$ **c** $4x^2 + 8x + 5$ **d** $2x^2 - 3x + 15$

4. **a** -44 **b** 25 **c** -80

5. 3

6. **a** -210 **b** $(x - 2)(2x + 3)(3x - 5)$

7.
- **a.** $k = 15$
- **b.** $(x + 3)(2x^2 - 3x + 5)$
- **c.** One solution $x = -3$ when $x + 3 = 0$.
  For $2x^2 - 3x + 5 = 0$, $b^2 - 4ac = -31 < 0$,
  so no other real solutions.

8. **a** 43 **b** $(x - 2)(3x^2 + 16x + 24) + 43$

9. **a** $(x - 1)(10x^2 - 19x - 15)$ **b** $(x - 1)(2x - 5)(5x + 3)$
c $x = 1$, $x = \frac{5}{2}$, $x = -\frac{3}{5}$

7.3 The factor theorem

1. **a** $f(-1) = 0$ **b** $f(-2) = 0$ **c** $f(3) = 0$ **d** $f(4) = 0$

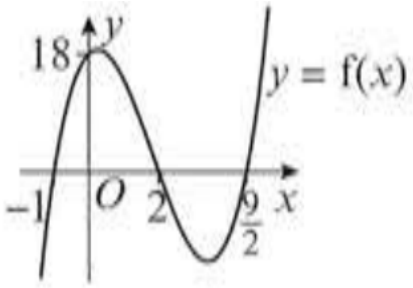
2.
- **a.** $f(2) = 0$, $(x - 2)(x + 3)(2x - 1)$
- **b.** $f(-3) = 0$, $(x + 3)(x + 5)(2x + 1)$
- **c.** $f(1) = 0$, $(x - 1)(3x - 2)(2x + 3)$
- **d.** $f(-4) = 0$, $(x + 4)(5x - 3)(3x + 2)$

3. **a** $(x - 1)(x - 3)(x + 6)$ **b** $(x + 2)(x + 5)(2x - 1)$
c $(x + 3)(x - 4)(3x + 5)$

4.

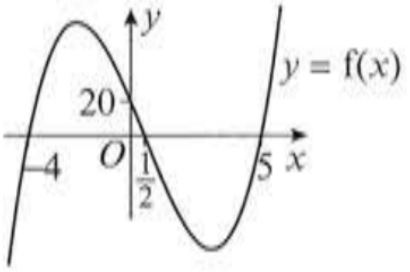
a. **i** $(x-2)(x+1)(2x-9)$

ii



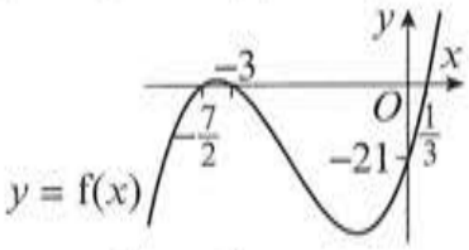
b. **i** $(x-5)(x+4)(2x-1)$

ii



c. **i** $(x+3)(2x+7)(3x-1)$

ii



5. $x = 3$, $x = -\frac{3}{2}$, $x = \frac{4}{3}$

6. **a** $f(-4) = 0$ **b** $(x+4)(3x-2)^2$

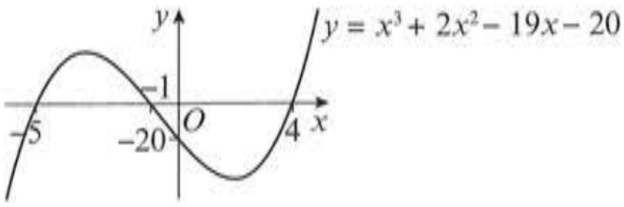
7. $(x-1)(x-2)(2x+3)$

8.

a. $g(-1) = -1 + 2 + 19 + k = 0 \Rightarrow k = -20$

b. $(x+1)(x-4)(x+5)$

c.



9. **a** $p(-3) = 0$ **b** $(x+3)(5x-2)^2$

c $x = -3$ and $x = \frac{2}{5}$

7.4 Mathematical proof

1. LHS $\equiv (x-4)(2x^2+15x+18) \equiv 2x^3+7x^2-42x-72 \equiv$ RHS

2. Using Pythagoras' theorem:
$PQ^2 = 16^2+12^2 = 400$, $QR^2 = 18^2+6^2 = 360$
$PR^2 = 2^2+6^2 = 40$
$QR^2+PR^2 = 360+40 = 400 = PQ^2$
As $PQ^2 = QR^2+PR^2$ using Pythagoras, triangle PQR is right-angled.

3. At point of intersection, $kx = 4x^2-5x+4$, so
$4x^2-x(5+k)+4=0$
Only one real root, so $b^2-4ac=0$ gives $k^2+10k-39=0$
$\Rightarrow (k+13)(k-3)=0 \Rightarrow k=3$ and $k=-13$, so the lines with equations $y=3x$ and $y=-13x$ are tangents to the curve.

4. If A, B and C lie on the same straight line, the gradients of the line segments AB, BC and AC are equal.
Gradient of $AB = \frac{1-3}{2+2} = -\frac{2}{4} = -\frac{1}{2}$
Gradient of $BC = \frac{-5-1}{14-2} = -\frac{6}{12} = -\frac{1}{2}$
As gradient of AB = gradient of BC and B is a common point, then A, B and C are collinear.

5. Completing the square,
$x^2-6x+10 = (x-3)^2-9+10 = (x-3)^2+1$
As $(x-3)^2 \geqslant 0$ for all values of x, then $(x-3)^2+1 \geqslant 1$
and $x^2-6x+10 > 0$ for all values of x.

6. LHS $= x^2+2xy+y^2-(x^2-2xy+y^2) = 4xy =$ RHS

7. If first odd number is $2n+1$, next consecutive odd is $2n+3$.

$$\begin{aligned}(2n+3)^2-(2n+1)^2 &= 4n^2+12n+9-(4n^2+4n+1)\\ &= 8n+8 = 8(n+1)\end{aligned}$$

So the difference between the squares of two consecutive odd numbers is a multiple of 8.

8. Let odd number be $2n+1$.

$$\begin{aligned}(2n+1)^3-(2n+1)^2 &= 8n^3+12n^2+6n+1-(4n^2+4n+1)\\ &= 8n^3+8n^2+2n\end{aligned}$$

As $8n^3+8n^2+2n = 2(4n^3+4n^2+n)$, the difference between the cube and the square of an odd number is even.

9. No real roots means $b^2-4ac<0$.
Solving $(k-3)^2-4\times 1\times(3-2k)=0$ gives
$k^2+2k-3=0 \Rightarrow (k+3)(k-1)=0$
Critical values are -3 and 1, so for no real roots $-3<k<1$.

7.5 Methods of proof

1.

a. $n = 5$: $5^2 + 5 + 17 = 47$; $n = 6$: $6^2 + 6 + 17 = 59$
$n = 7$: $7^2 + 7 + 17 = 73$; $n = 8$: $8^2 + 8 + 17 = 89$
$n = 9$: $9^2 + 9 + 17 = 107$; $n = 10$: $10^2 + 10 + 17 = 127$
Proof by exhaustion, so for $5 \leqslant n \leqslant 10$, $f(n)$ is prime.

b. When $n = 17$: $17^2 + 17 + 17 = 17(17 + 1 + 1)$, divisible by 17 so not prime. Counter-example, so $f(n)$ is not prime for all positive integers, n.

2. 2 and 3 are both prime numbers. $3 - 2 = 1$ is not even, so the difference of two prime numbers is not always even.

3.

a. $(1 + 2x)^2 - 4x^2 - 1 = 4x > 0$ for all positive values of x.
So $(1 + 2x)^2 - 4x^2 - 1 > 0 \Rightarrow (1 + 2x)^2 > 1 + 4x^2$

b. If $x = -1$, $(1 + 2x)^2 = 1$ and $1 + 4x^2 = 5$, counter-example, so $(1 + 2x)^2 > 1 + 4x^2$ is not true for all values of x.

4. **a** Cancelling a factor of x is not valid when $x = 0$.
b $x = 0$

5. When $x = -2$, $(-2 + 2)^2 = 0$ and $(-2)^2 = 4$, so $(x + 2)^2 > x^2$ is not true when $x = -2$.

6. $n = 1$: $9 > 4$; $n = 2$: $25 > 8$; $n = 3$: $49 > 16$; $n = 4$: $81 > 32$
$n = 5$: $121 > 64$; $n = 6$: $169 > 128$

7. Sometimes true. True for all values except when $0 < x < 1$.

8. $(x + 5)^2 - 4x - 9 = x^2 + 6x + 16 = (x + 3)^2 + 7 \geqslant 0$ for all real values of x, proved by completing the square.
So $(x + 5)^2 - 4x - 9 \geqslant 0 \Rightarrow (x + 5)^2 \geqslant 4x + 9$

9. Proof by exhaustion:
$n = 2$: $2^2 + 2 = 6$ not divisible by 4
$n = 3$: $3^2 + 2 = 11$ not divisible by 4
$n = 4$: $4^2 + 2 = 18$ not divisible by 4
$n = 5$: $5^2 + 2 = 27$ not divisible by 4
$n = 6$: $6^2 + 2 = 38$ not divisible by 4
$n = 7$: $7^2 + 2 = 51$ not divisible by 4

Problem solving: Set A

Bronze

a. -24

b. $(x - 4)(x^2 - 5x - 18) - 24$

c. $(x + 2)(x - 3)(x - 8)$

d. Proof by exhaustion:
$n = 4$: $p(4) = 4^3 - 9 \times 4^2 + 2 \times 4 + 48 = -24 < 0$
$n = 5$: $p(5) = 5^3 - 9 \times 5^2 + 2 \times 5 + 48 = -42 < 0$
$n = 6$: $p(6) = 6^3 - 9 \times 6^2 + 2 \times 6 + 48 = -48 < 0$
$n = 7$: $p(7) = 7^3 - 9 \times 7^2 + 2 \times 7 + 48 = -36 < 0$
So for $3 < n < 8$, $n^3 - 9n^2 + 2n + 48 < 0$

Silver

a. $(2x+1)(x^2-8x+20)$

b. One root when $2x+1=0$, i.e. $x=-\frac{1}{2}$.
Using discriminant for $x^2-8x+20$,
$b^2-4ac=64-4\times1\times20=-16<0$ indicates no real roots so $p(x)=0$ has exactly one real root.

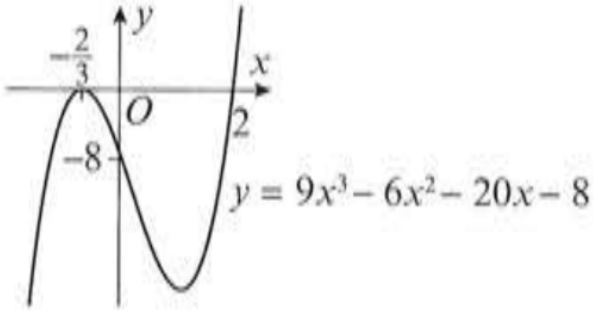
c. $2x^3-15x^2+32x+20=(2x+1)(x^2-8x+20)$
$2x+1>0$ when $x>-\frac{1}{2}$, so $2x+1>0$ for all real positive values of x.
Completing the square for $x^2-8x+20$ gives $(x-4)^2+4$. As $(x-4)^2\geqslant 0$ for all real values of x, $(x-4)^2+4\geqslant 4$ for all real values of x and $(2x+1)(x^2-8x+20)>0$ for all real positive values of x.

Gold

a. $a=9$, $b=12$, $c=4$, $d=-2$

b. $p(x)=(x-2)(9x^2+12x+4)$. One real root when $x=2$.
Using discriminant for $9x^2+12x+4$,
$b^2-4ac=144-4\times9\times4=0$ indicates one real root, so $p(x)=0$ has exactly two real roots.

c.



d. $\{x : x<-\frac{2}{3}\}\cup\{x : -\frac{2}{3}<x<2\}$

Problem solving: Set B

Bronze

a. $f(-1)=-1-1-p+q=0$, which gives $q-p=2$

b. $f(-3)=-27-9-3p+q=0$, which gives $q-3p=36$

c. $p=-17$ and $q=-15$

d. $f(x)=(x+1)(x+3)(x-5)$

Silver

a. $f(-2) = -16 - 4 - 2p + q = 0$, gives $q - 2p - 20 = 0$

b. $p = -13$ and $q = -6$

c. $f(x) = (x + 2)(x - 3)(2x + 1)$

Gold

a. $f(2) = 8 + (p + 4) \times 4 + 16 + q = 0$, which gives $4p + q + 40 = 0$

b. $f(-p) = -p^3 + p^3 + 4p^2 - 8p + q = 0$.
Simplify to obtain $4p^2 - 8p + q = 0$

c. $p = 5$ and $q = -60$

d. $f(x) = (x - 2)(x + 5)(x + 6)$

Selected exam question bank answers

Question 30

a. Consider $(p - q)^2 = p^2 + q^2 - 2pq = p^2 - q^2 + 2q^2 - 2pq$
If $p > q > 0$ then $2pq > 2q^2$, so $2q^2 - 2pq$ is negative, and
$(p - q)^2 < p^2 - q^2$, so $\sqrt{p^2 - q^2} > p - q$ as required.

b. $p = 0$, $q = 0$ would give $0 > 0$ which is untrue.

Question 34

a. When $x = -1$, $5^x = 5^{-1} = \frac{1}{5}$; $2^x = 2^{-1} = \frac{1}{2}$ and $\frac{1}{5} < \frac{1}{2}$, so not true for all values of x.

b. Write the consecutive odd numbers as $2n - 1$ and $2n + 1$, where n is an integer. Then:

$$\begin{aligned}\tfrac{1}{2}(2n-1)^2 + \tfrac{1}{2}(2n+1)^2 &= \tfrac{1}{2}(4n^2 - 4n + 1) + \tfrac{1}{2}(4n^2 + 4n + 1) \\ &= 2n^2 - 2n + \tfrac{1}{2} + 2n^2 + 2n + \tfrac{1}{2} \\ &= 4n^2 + 1\end{aligned}$$

which is one more than an even number, so must be odd.

Question 41

a. $(x + 1)(2x + 1)(3x - 2)$

b. $\dfrac{6x^3 + 5x^2 - 3x - 2}{2x^3 + 3x^2 + x} = \dfrac{(x + 1)(2x + 1)(3x - 2)}{x(2x + 1)(x + 1)} = 3 - \dfrac{2}{x}$

Question 57

a. Factor theorem states if $(x+3)$ is a factor of $g(x)$, then $g(-3)=0$.
$g(-3) = 4\times(-3)^3 - 16\times(-3)^2 - 35\times(-3) + 147$
$= -108 - 144 + 105 + 147 = 0$
so $(x+3)$ is a factor of $g(x)$.

b. By division $g(x) = (x+3)(4x^2 - 28x + 49)$
$= (x+3)(2x-7)^2$

c. **i** $g(x) \leqslant 0$ when $x \leqslant -3$
ii $g(2x) \leqslant 0$ when $x \leqslant -\frac{3}{2}$

Question 69

a. 7

b. $f(1) = (1+k)(2+5) + 7 = 0$, $7 + 7k + 7 = 0$ so $7k = -14$ and $k = -2$

c. $x = -\frac{1}{2}$, $x = -3$ and $x = 1$

Question 75

a. $f(5) = 250 + 25k - 120 + 45 = 0$ so $25k = -175$ and $k = -7$

b. $f(x) = (x-5)(2x-3)(x+3)$

c. $5^y = 5$ so $y = 1$.
$5^y = \frac{3}{2} \Rightarrow y\ln 5 = \ln\frac{3}{2}$ so $y = 0.25$
($5^y = -3$ has no solutions)

Question 83

a. $f(-2) = 0$

b. $f(x) = (x+2)(x^2 - 4x + 9)$; using discriminant on quadratic, $b^2 - 4ac = 16 - 36 = -20 < 0$ so no real roots.
Only one root when $x + 2 = 0$, i.e. $x = -2$

c. $x^3 - 2x^2 + x + 18 \geqslant 18$, means $x^3 - 2x^2 + x \geqslant 0$.
Factorising $x(x^2 - 2x + 1) = x(x-1)^2$, $(x-1)^2 \geqslant 0$ for all values of x. If $x \geqslant 0$ then $x(x-1)^2 \geqslant 0$ and
$f(x) = x^3 - 2x^2 + x + 18 = x(x-1)^2 + 18 \geqslant 18$ for $x \geqslant 0$.